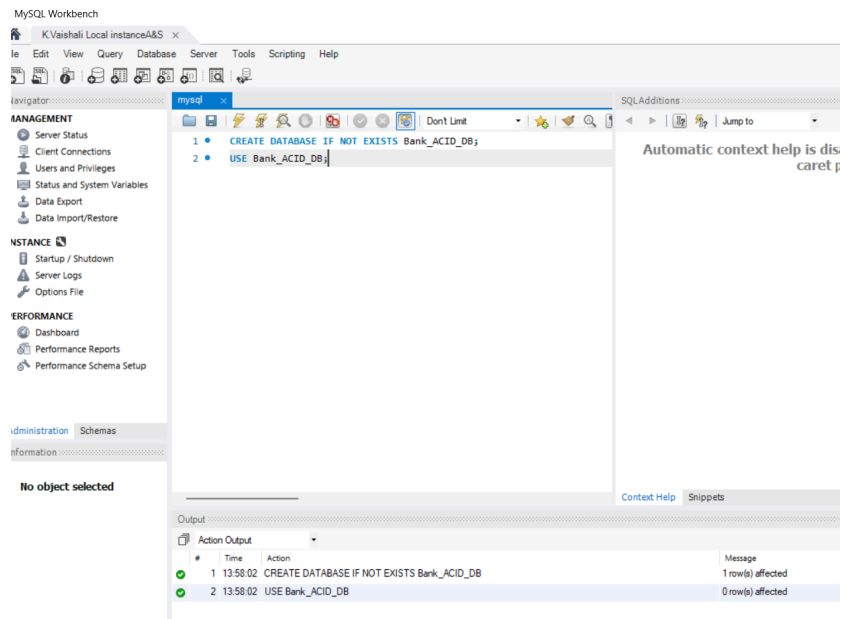


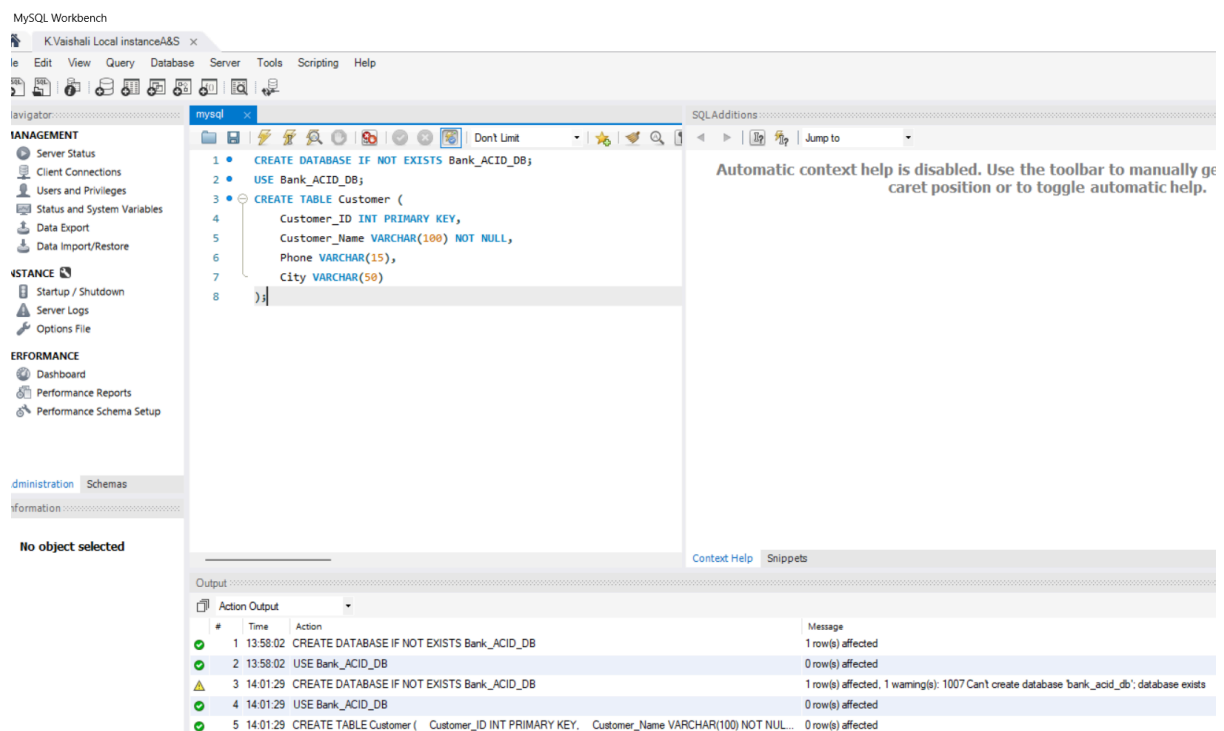
BANK DATABASE - ACID PROPERTIES & ISOLATION

Name-K.Vaishali
RollNo-2520030117
Section-7
subject-DBS&DBE

STEP-1 CREATE DATABASE



STEPS-2 CREATE CUSTOMER TABLE



STEP-3 CREATE ACCOUNT TABLE

MySQL Workbench

K Vaishali Local instanceA&S

File Edit View Query Database Server Tools Scripting Help

Navigator: mysql

ANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

```
1 CREATE DATABASE IF NOT EXISTS Bank_ACID_DB;
2 USE Bank_ACID_DB;
3 CREATE TABLE Customer (
4     Customer_ID INT PRIMARY KEY,
5     Customer_Name VARCHAR(100) NOT NULL,
6     Phone VARCHAR(15),
7     City VARCHAR(50)
8 );
9 CREATE TABLE Account (
10     Account_No INT PRIMARY KEY,
11     Customer_ID INT,
12     Account_Type VARCHAR(20),
13     Balance DECIMAL(12,2),
14     Branch VARCHAR(50),
15
16     FOREIGN KEY (Customer_ID)
17     REFERENCES Customer(Customer_ID)
18 );
```

Automatic context help is disabled. Use the toolbar to manually get caret position or to toggle automatic help.

Context Help Snippets

Output

#	Time	Action	Message
3	14:01:29	CREATE DATABASE IF NOT EXISTS Bank_ACID_DB	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bank_acid_db'; database exists
4	14:01:29	USE Bank_ACID_DB	0 row(s) affected
5	14:01:29	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL, Phone VARCHAR(15), City VARCHAR(50))	0 row(s) affected
6	14:02:43	CREATE DATABASE IF NOT EXISTS Bank_ACID_DB	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bank_acid_db'; database exists
7	14:02:43	USE Bank_ACID_DB	0 row(s) affected
8	14:02:43	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL, Phone VARCHAR(15), City VARCHAR(50))	Error Code: 1050. Table 'customer' already exists

Object Info Session

STEP-4 CREATE TRANSACTION TABLE

MySQL Workbench

K Vaishali Local instanceA&S

File Edit View Query Database Server Tools Scripting Help

Navigator: mysql

ANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

```
9 CREATE TABLE Account (
10     Account_No INT PRIMARY KEY,
11     Customer_ID INT,
12     Account_Type VARCHAR(20),
13     Balance DECIMAL(12,2),
14     Branch VARCHAR(50),
15
16     FOREIGN KEY (Customer_ID)
17     REFERENCES Customer(Customer_ID)
18 );
19 CREATE TABLE Bank_Transaction (
20     Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
21     Account_No INT,
22     Transaction_Type VARCHAR(20),
23     Amount DECIMAL(12,2),
24     Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
25
26     FOREIGN KEY (Account_No)
27     REFERENCES Account(Account_No)
28 );
```

Automatic context help is disabled. Use the toolbar to manually get caret position or to toggle automatic help.

Context Help Snippets

Output

#	Time	Action	Message
9	14:05:04	CREATE DATABASE IF NOT EXISTS Bank_ACID_DB	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bank_acid_db'; database exists
10	14:05:04	USE Bank_ACID_DB	0 row(s) affected
11	14:05:04	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL, Phone VARCHAR(15), City VARCHAR(50))	Error Code: 1050. Table 'customer' already exists
12	14:06:26	CREATE DATABASE IF NOT EXISTS Bank_ACID_DB	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bank_acid_db'; database exists
13	14:06:26	USE Bank_ACID_DB	0 row(s) affected
14	14:06:26	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL, Phone VARCHAR(15), City VARCHAR(50))	Error Code: 1050. Table 'customer' already exists

Object Info Session

STEP-5 INSERT CUSTOMER DATA

K.Vaishali Local instanceA&S x

Edit View Query Database Server Tools Scripting Help

mysql

```
17 REFERENCES Customer(Customer_ID)
18 );
19 CREATE TABLE Bank_Transaction (
20 Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
21 Account_No INT,
22 Transaction_Type VARCHAR(20),
23 Amount DECIMAL(12,2),
24 Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
25
26 FOREIGN KEY (Account_No)
27 REFERENCES Account(Account_No)
28 );
29 INSERT INTO Customer
30 (Customer_ID, Customer_Name, Phone, City)
31 VALUES
32 (101, 'K.Vaishali', '9876543210', 'Hyderabad'),
33 (102, 'K.Vaishali', '9876543211', 'Vijayawada'),
34 (103, 'Arjun Reddy', '9876543212', 'Bangalore'),
35 (104, 'Sneha Rao', '9876543213', 'Chennai'),
36 (105, 'Kiran Kumar', '9876543214', 'Hyderabad');
```

Automatic context help is disabled. Use the toolbar to manually get help or to toggle automatic help.

Output

#	Time	Action	Message
22	14:17:24	USE Bank_ACID_DB	0 row(s) affected
23	14:17:24	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL...	Error Code: 1050. Table 'customer' already exists
24	14:17:37	CREATE DATABASE IF NOT EXISTS Bank_ACID_DB	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bank_acid_db'; database exists
25	14:17:37	USE Bank_ACID_DB	0 row(s) affected
26	14:17:37	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL...	Error Code: 1050. Table 'customer' already exists

STEP-6 INSERT ACCOUNT DATA

MySQL Workbench

K.Vaishali Local instanceA&S x

Edit View Query Database Server Tools Scripting Help

mysql

```
30 FOREIGN KEY (Account_No)
31 REFERENCES Account(Account_No)
32 );
33
34 INSERT INTO Customer
35 (Customer_ID, Customer_Name, Phone, City)
36 VALUES
37 (101, 'K.Vaishali', '9876543210', 'Hyderabad'),
38 (102, 'K.Vaishali', '9876543211', 'Vijayawada'),
39 (103, 'Arjun Reddy', '9876543212', 'Bangalore'),
40 (104, 'Sneha Rao', '9876543213', 'Chennai'),
41 (105, 'Kiran Kumar', '9876543214', 'Hyderabad');
42 INSERT INTO Account
43 (Account_No, Customer_ID, Account_Type, Balance, Branch)
44 VALUES
45 (10001, 101, 'Savings', 50000, 'Hyderabad'),
46 (10002, 102, 'Savings', 75000, 'Vijayawada'),
47 (10003, 103, 'Current', 120000, 'Bangalore'),
48 (10004, 104, 'Savings', 45000, 'Chennai'),
49 (10005, 105, 'Current', 90000, 'Hyderabad');
```

Automatic context help is disabled. Use the toolbar to manually get help or to toggle automatic help.

Output

#	Time	Action	Message
16	14:28:39	CREATE DATABASE IF NOT EXISTS Bank_ACID_DB	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bank_acid_db'; database exists
17	14:28:39	USE Bank_ACID_DB	0 row(s) affected
18	14:28:39	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL...	Error Code: 1050. Table 'customer' already exists
19	14:29:01	CREATE DATABASE IF NOT EXISTS Bank_ACID_DB	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bank_acid_db'; database exists
20	14:29:01	USE Bank_ACID_DB	0 row(s) affected
21	14:29:01	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NU...	Error Code: 1050. Table 'customer' already exists

STEP-7 INSERT BANK TRANSACTION DATA

MySQL Workbench

K.Vaishali Local instanceA&S

Edit View Query Database Server Tools Scripting Help

mysql

Automatic context help is disabled. Use the toolbar to manually
caret position or to toggle automatic help

```
38 (102, 'K.Vaishali', '9876543211', 'Vijayawada'),
39 (103, 'Arjun Reddy', '9876543212', 'Bangalore'),
40 (104, 'Sneha Rao', '9876543213', 'Chennai'),
41 (105, 'Kiran Kumar', '9876543214', 'Hyderabad');
42 • INSERT INTO Account
43 (Account_No, Customer_ID, Account_Type, Balance, Branch)
44 VALUES
45 (10001, 101, 'Savings', 50000, 'Hyderabad'),
46 (10002, 102, 'Savings', 75000, 'Vijayawada'),
47 (10003, 103, 'Current', 120000, 'Bangalore'),
48 (10004, 104, 'Savings', 45000, 'Chennai'),
49 (10005, 105, 'Current', 90000, 'Hyderabad');
50 • INSERT INTO Bank_Transaction
51 (Account_No, Transaction_Type, Amount)
52 VALUES
53 (10001, 'DEPOSIT', 10000),
54 (10002, 'DEPOSIT', 15000),
55 (10003, 'WITHDRAW', 20000),
56 (10004, 'DEPOSIT', 5000),
57 (10005, 'WITHDRAW', 10000);
```

Output

#	Time	Action	Message
19	14:29:01	CREATE DATABASE IF NOT EXISTS Bank_ACID_DB	1 row(s) affected. 1 warning(s): 1007 Can't create database 'bank_acid_db'; database exists
20	14:29:01	USE Bank_ACID_DB	0 row(s) affected
21	14:29:01	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NU...	Error Code: 1050. Table 'customer' already exists
22	14:30:53	CREATE DATABASE IF NOT EXISTS Bank_ACID_DB	1 row(s) affected. 1 warning(s): 1007 Can't create database 'bank_acid_db'; database exists
23	14:30:53	USE Bank_ACID_DB	0 row(s) affected
24	14:30:53	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NU...	Error Code: 1050. Table 'customer' already exists

STEP-8 DISPLAY DATA

MySQL Workbench

K.Vaishali Local instanceA&S

Edit View Query Database Server Tools Scripting Help

mysql

Automatic context help is disabled. Use the t
caret position or to tog

```
49 • INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount)
50 VALUES
51 (10001, 'DEPOSIT', 10000),
52 (10002, 'DEPOSIT', 15000),
53 (10003, 'WITHDRAW', 20000),
54 (10004, 'DEPOSIT', 5000),
55 (10005, 'WITHDRAW', 10000);
56 • SELECT * FROM Customer;
57 • SELECT * FROM Account;
```

Result Grid

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	50000.00	Hyderabad
	10002	102	Savings	75000.00	Vijayawada
	10003	103	Current	120000.00	Bangalore
	10004	104	Savings	45000.00	Chennai
	10005	105	Current	90000.00	Hyderabad
*	NULL	NULL	NULL	NULL	NULL

Customer 4 Account 5

Output

#	Time	Action	Message
56	14:34:44	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account...	0 row(s) affected
57	14:34:44	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, City) VALUES (101, 'K.Vaishali', '987654321...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
58	14:34:44	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, '...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
59	14:34:44	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
60	14:34:44	SELECT * FROM Customer	5 row(s) returned
61	14:34:44	SELECT * FROM Account	5 row(s) returned

K Vaishali Local instanceA&S x

File Edit View Query Database Server Tools Scripting Help

mysql x

SQL Additions: Jump to

Automatic context help is disabled. Use the toolbar caret position or to toggle

```

50 VALUES
51 (10001, 'DEPOSIT', 10000),
52 (10002, 'DEPOSIT', 15000),
53 (10003, 'WITHDRAW', 20000),
54 (10004, 'DEPOSIT', 5000),
55 (10005, 'WITHDRAW', 10000);
56 SELECT * FROM Customer;
57 SELECT * FROM Account;
58 SELECT * FROM Bank_Transaction;

```

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-28 14:33:21
2	10002	DEPOSIT	15000.00	2026-08-28 14:33:21
3	10003	WITHDRAW	20000.00	2026-08-28 14:33:21
4	10004	DEPOSIT	5000.00	2026-08-28 14:33:21
5	10005	WITHDRAW	10000.00	2026-08-28 14:33:21
NULL	NULL	NULL	NULL	NULL

Customer 1 Account 2 Bank_Transaction 3 x Apply Revert Context Help Snippets

Output: Action Output

#	Time	Action	Message
43	14:33:21	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, City) VALUES (101, 'K.Vaishali', '987654321...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
44	14:33:21	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
45	14:33:21	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
46	14:33:21	SELECT * FROM Customer	5 row(s) returned
47	14:33:21	SELECT * FROM Account	5 row(s) returned
48	14:33:21	SELECT * FROM Bank_Transaction	5 row(s) returned

Info Session

STEP-9 START A TRANSACTION

K Vaishali Local instanceA&S x

File Edit View Query Database Server Tools Scripting Help

mysql x

SQL Additions: Jump to

Automatic context help is disabled. Use the toolbar caret position or to toggle

```

50 VALUES
51 (10001, 'DEPOSIT', 10000),
52 (10002, 'DEPOSIT', 15000),
53 (10003, 'WITHDRAW', 20000),
54 (10004, 'DEPOSIT', 5000),
55 (10005, 'WITHDRAW', 10000);
56 SELECT * FROM Customer;
57 SELECT * FROM Account;
58 START TRANSACTION;

```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Customer 6 Account 7 x Apply Revert Context Help Snippets

Output: Action Output

#	Time	Action	Message
70	14:38:17	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, City) VALUES (101, 'K.Vaishali', '987654321...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
71	14:38:17	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
72	14:38:17	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
73	14:38:17	SELECT * FROM Customer	5 row(s) returned
74	14:38:17	SELECT * FROM Account	5 row(s) returned
75	14:38:17	START TRANSACTION	0 row(s) affected

Info Session

STEP-10 DEPOSIT MONEY

The screenshot shows the MySQL Workbench interface with the following components:

- Left Sidebar (Navigation):** Includes sections for GEMENT (Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, Data Import/Restore), NCE (Startup / Shutdown, Server Logs, Options File), and RMANCE (Dashboard, Performance Reports, Performance Schema Setup).
- SQL Editor:** Contains the following SQL script:

```
42 VALUES
43 (10001, 101, 'Savings', 50000, 'Hyderabad'),
44 (10002, 102, 'Savings', 75000, 'Vijayawada'),
45 (10003, 103, 'Current', 120000, 'Bangalore'),
46 (10004, 104, 'Savings', 45000, 'Chennai'),
47 (10005, 105, 'Current', 90000, 'Hyderabad');
48
49 • INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount
50 VALUES
51 (10001, 'DEPOSIT', 10000),
52 (10002, 'DEPOSIT', 15000),
53 (10003, 'WITHDRAW', 20000),
54 (10004, 'DEPOSIT', 5000),
55 (10005, 'WITHDRAW', 10000);
56 • SELECT * FROM Customer;
57 • SELECT * FROM Account;
58 • START TRANSACTION;
59 • UPDATE Account
60 SET Balance = Balance + 5000
61 WHERE Account_No = 10001;
```
- Output Panel:** Shows the execution results of the SQL script:

#	Time	Action	Message
85	14:39:46	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
86	14:39:46	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
87	14:39:46	SELECT * FROM Customer	5 row(s) returned
88	14:39:46	SELECT * FROM Account	5 row(s) returned
89	14:39:46	START TRANSACTION	0 row(s) affected
90	14:39:46	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
- Right Panel:** Displays a message: "Automatic context help is disabled. Use the tool caret position or to toggle".

STEP-11 CHECK BALANCE

The screenshot shows the MySQL Workbench interface with the following components:

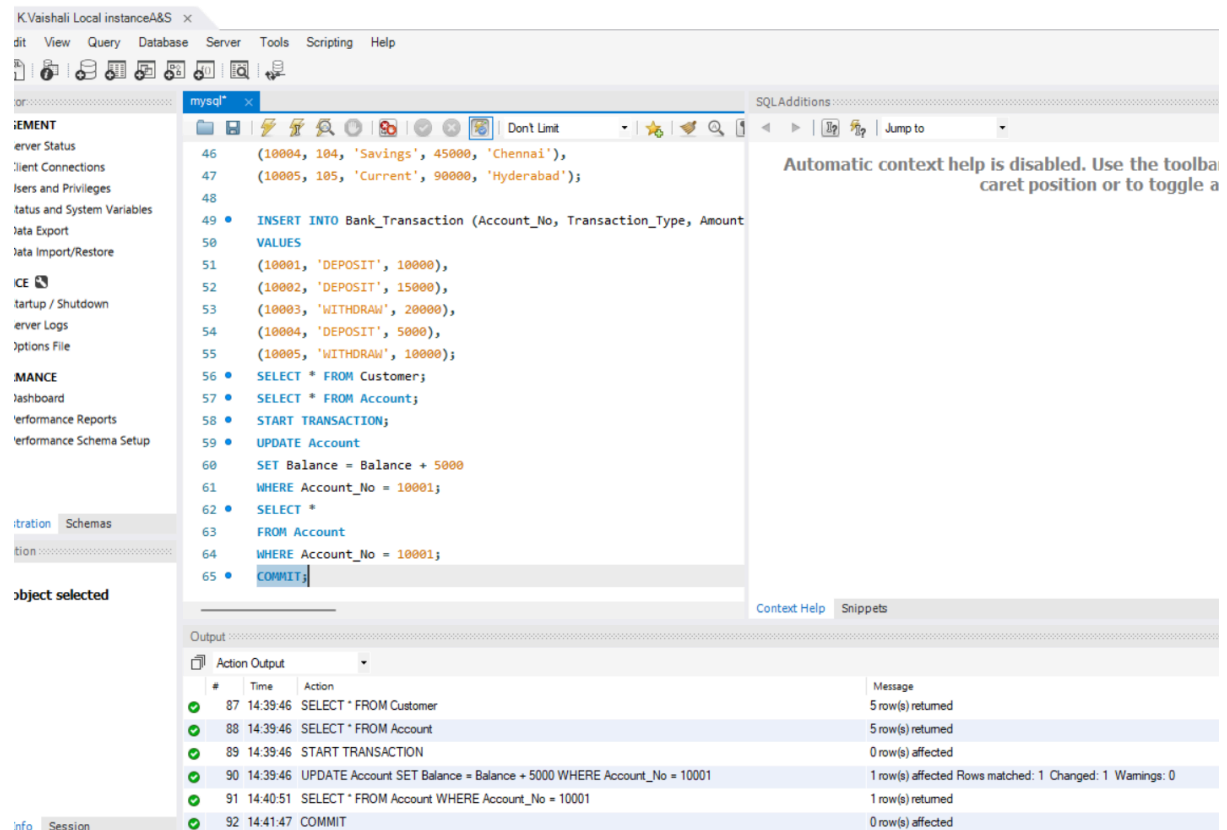
- Left Sidebar (Navigation):** Same as Step 10.
- SQL Editor:** Contains the following SQL script:

```
56 • SELECT * FROM Customer;
57 • SELECT * FROM Account;
58 • START TRANSACTION;
59 • UPDATE Account
60 SET Balance = Balance + 5000
61 WHERE Account_No = 10001;
62 • SELECT *
63 FROM Account
64 WHERE Account_No = 10001;
```
- Result Grid:** Displays the results of the SQL script:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad
- Output Panel:** Shows the execution results of the SQL script:

#	Time	Action	Message
86	14:39:46	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
87	14:39:46	SELECT * FROM Customer	5 row(s) returned
88	14:39:46	SELECT * FROM Account	5 row(s) returned
89	14:39:46	START TRANSACTION	0 row(s) affected
90	14:39:46	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
91	14:40:51	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
- Right Panel:** Displays a message: "Automatic context help is disabled. Use the tool caret position or to toggle".

STEP-12 COMMIT TRANSACTION

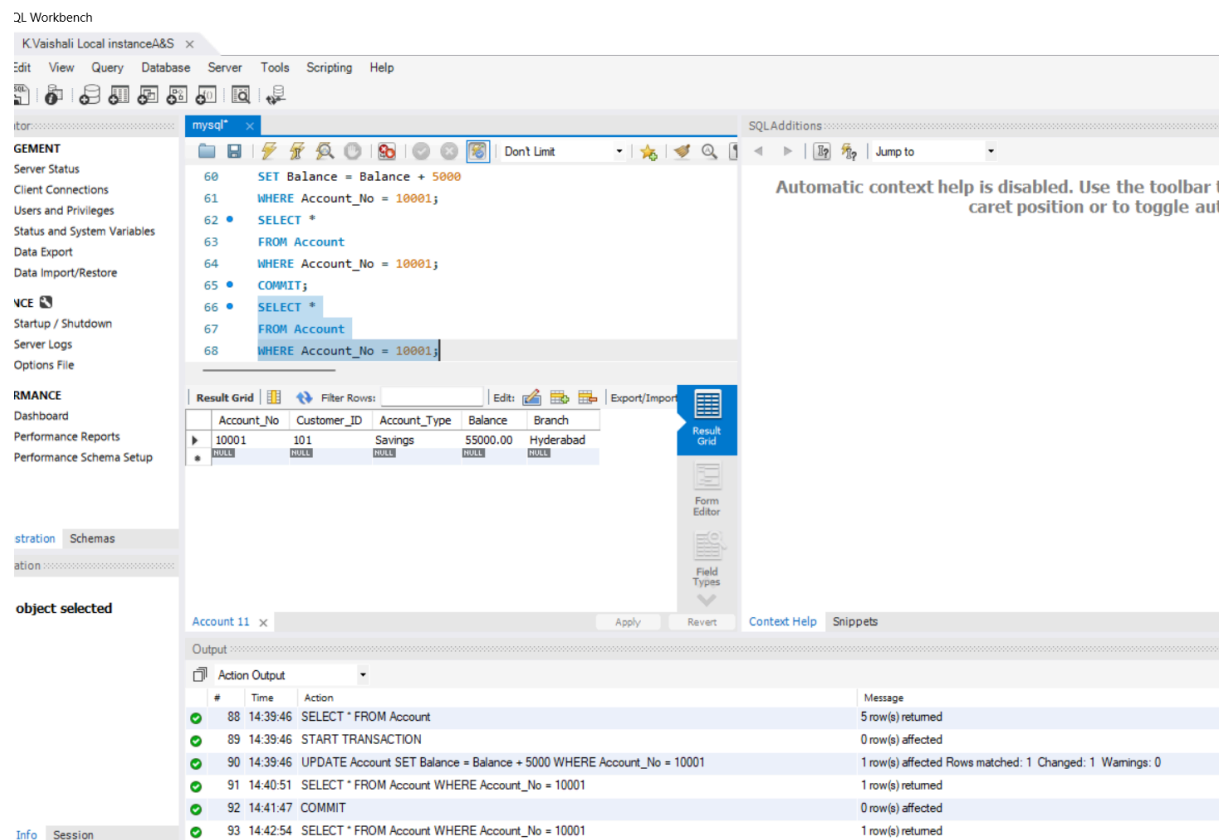


Automatic context help is disabled. Use the toolbar caret position or to toggle a

```
46 (10004, 104, 'Savings', 45000, 'Chennai');
47 (10005, 105, 'Current', 90000, 'Hyderabad');
48
49 • INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount
50 VALUES
51 (10001, 'DEPOSIT', 10000),
52 (10002, 'DEPOSIT', 15000),
53 (10003, 'WITHDRAW', 20000),
54 (10004, 'DEPOSIT', 5000),
55 (10005, 'WITHDRAW', 10000);
56 • SELECT * FROM Customer;
57 • SELECT * FROM Account;
58 • START TRANSACTION;
59 • UPDATE Account
60 SET Balance = Balance + 5000
61 WHERE Account_No = 10001;
62 • SELECT *
63 FROM Account
64 WHERE Account_No = 10001;
65 • COMMIT;
```

#	Time	Action	Message
✓ 87	14:39:46	SELECT * FROM Customer	5 row(s) returned
✓ 88	14:39:46	SELECT * FROM Account	5 row(s) returned
✓ 89	14:39:46	START TRANSACTION	0 row(s) affected
✓ 90	14:39:46	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 91	14:40:51	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
✓ 92	14:41:47	COMMIT	0 row(s) affected

STEP-13 CHECK BALANCE AFTER COMMIT



Automatic context help is disabled. Use the toolbar caret position or to toggle aul

```
60 SET Balance = Balance + 5000
61 WHERE Account_No = 10001;
62 • SELECT *
63 FROM Account
64 WHERE Account_No = 10001;
65 • COMMIT;
66 • SELECT *
67 FROM Account
68 WHERE Account_No = 10001;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad

#	Time	Action	Message
✓ 88	14:39:46	SELECT * FROM Account	5 row(s) returned
✓ 89	14:39:46	START TRANSACTION	0 row(s) affected
✓ 90	14:39:46	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 91	14:40:51	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
✓ 92	14:41:47	COMMIT	0 row(s) affected
✓ 93	14:42:54	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned

STEP-14 START TRANSACTION

MySQL Workbench interface showing a transaction being executed.

SQL Editor:

```
50 VALUES
51 (10001, 'DEPOSIT', 10000),
52 (10002, 'DEPOSIT', 15000),
53 (10003, 'WITHDRAW', 20000),
54 (10004, 'DEPOSIT', 5000),
55 (10005, 'WITHDRAW', 10000);
56 • SELECT * FROM Customer;
57 • SELECT * FROM Account;
58 • START TRANSACTION;
59 • UPDATE Account
60 SET Balance = Balance + 5000
61 WHERE Account_No = 10001;
62 • SELECT *
63 FROM Account
64 WHERE Account_No = 10001;
65 • COMMIT;
66 • SELECT *
67 FROM Account
68 WHERE Account_No = 10001;
69 • START TRANSACTION;
```

Output:

#	Time	Action	Message
✓ 107	14:43:47	START TRANSACTION	0 row(s) affected
✓ 108	14:43:47	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 109	14:43:47	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
✓ 110	14:43:47	COMMIT	0 row(s) affected
✓ 111	14:43:47	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
✓ 112	14:43:47	START TRANSACTION	0 row(s) affected

STEP-15 WITHDRAW MONEY

MySQL Workbench interface showing a transaction being executed.

SQL Editor:

```
53 (10003, 'WITHDRAW', 20000),
54 (10004, 'DEPOSIT', 5000),
55 (10005, 'WITHDRAW', 10000);
56 • SELECT * FROM Customer;
57 • SELECT * FROM Account;
58 • START TRANSACTION;
59 • UPDATE Account
60 SET Balance = Balance + 5000
61 WHERE Account_No = 10001;
62 • SELECT *
63 FROM Account
64 WHERE Account_No = 10001;
65 • COMMIT;
66 • SELECT *
67 FROM Account
68 WHERE Account_No = 10001;
69 • START TRANSACTION;
70 • UPDATE Account
71 SET Balance = Balance - 10000
72 WHERE Account_No = 10001;
```

Output:

#	Time	Action	Message
✓ 108	14:43:47	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 109	14:43:47	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
✓ 110	14:43:47	COMMIT	0 row(s) affected
✓ 111	14:43:47	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
✓ 112	14:43:47	START TRANSACTION	0 row(s) affected
✓ 113	14:46:26	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

STEP-16 CHECK BALANCE

/SQL Workbench

SQL Workbench interface showing a transaction for account 10001. The transaction includes a SELECT, START TRANSACTION, UPDATE (decreasing balance by 10000), and another SELECT. The output shows 1 row affected for the update and 1 row returned for the final SELECT.

SQL Query:

```
67 FROM Account
68 WHERE Account_No = 10001;
69 • START TRANSACTION;
70 • UPDATE Account
71 SET Balance = Balance - 10000
72 WHERE Account_No = 10001;
73 • SELECT *
74 FROM Account
75 WHERE Account_No = 10001;
```

Result Grid:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	45000.00	Hyderabad

Output:

#	Time	Action	Message
109	14:43:47	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
110	14:43:47	COMMIT	0 row(s) affected
111	14:43:47	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
112	14:43:47	START TRANSACTION	0 row(s) affected
113	14:46:26	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
114	14:47:55	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned

STEP-17 ROLLBACK

SQL Workbench

SQL Workbench interface showing a transaction for account 10001. The transaction includes a SELECT, START TRANSACTION, UPDATE (decreasing balance by 10000), a ROLLBACK, and a final SELECT. The output shows 0 rows affected for the update and 1 row returned for the final SELECT.

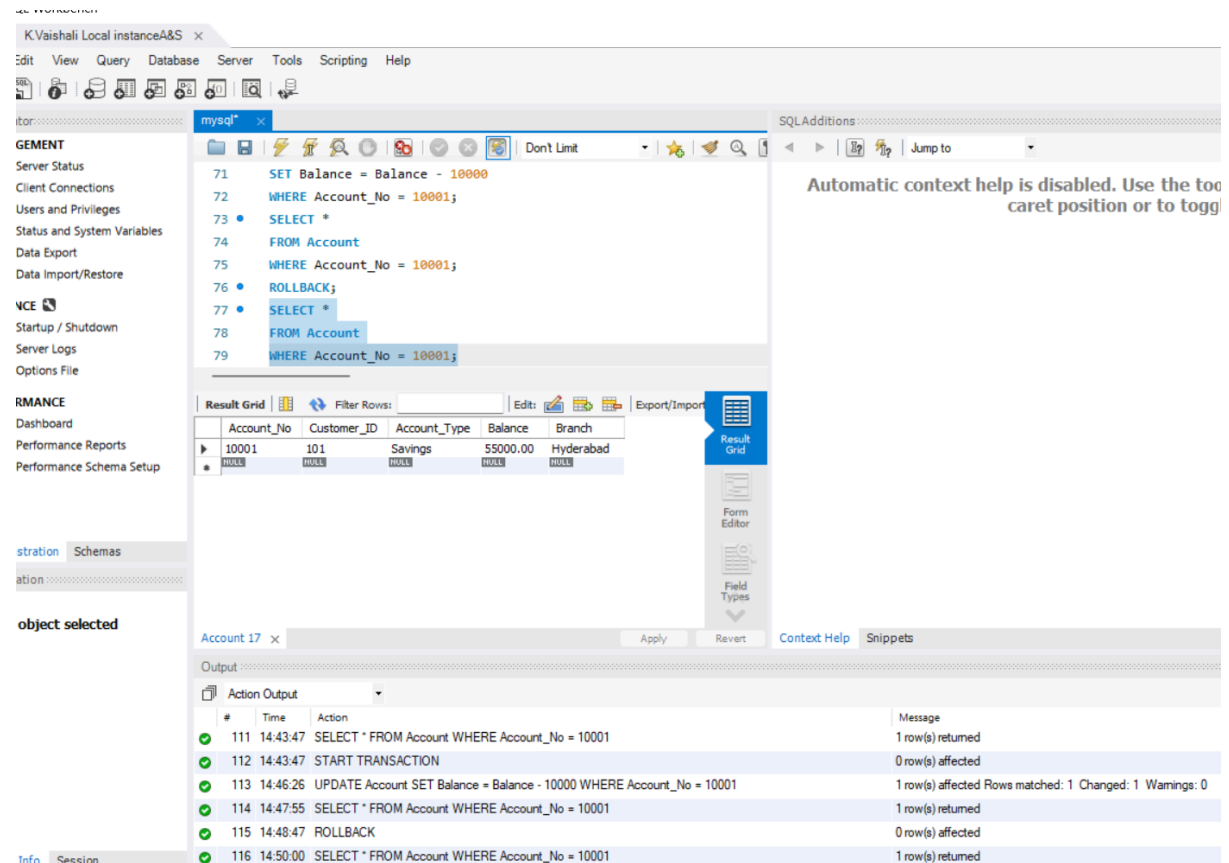
SQL Query:

```
57 • SELECT * FROM Account;
58 • START TRANSACTION;
59 • UPDATE Account
60 SET Balance = Balance + 5000
61 WHERE Account_No = 10001;
62 • SELECT *
63 FROM Account
64 WHERE Account_No = 10001;
65 • COMMIT;
66 • SELECT *
67 FROM Account
68 WHERE Account_No = 10001;
69 • START TRANSACTION;
70 • UPDATE Account
71 SET Balance = Balance - 10000
72 WHERE Account_No = 10001;
73 • SELECT *
74 FROM Account
75 WHERE Account_No = 10001;
76 • ROLLBACK;
```

Output:

#	Time	Action	Message
110	14:43:47	COMMIT	0 row(s) affected
111	14:43:47	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
112	14:43:47	START TRANSACTION	0 row(s) affected
113	14:46:26	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
114	14:47:55	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
115	14:48:47	ROLLBACK	0 row(s) affected

STEP-18 CHECK BALANCE AFTER ROLLBACK



The screenshot shows the MySQL Workbench interface. The query window contains the following SQL code:

```
71 SET Balance = Balance - 10000
72 WHERE Account_No = 10001;
73 SELECT *
74 FROM Account
75 WHERE Account_No = 10001;
76 ROLLBACK;
77 SELECT *
78 FROM Account
79 WHERE Account_No = 10001;
```

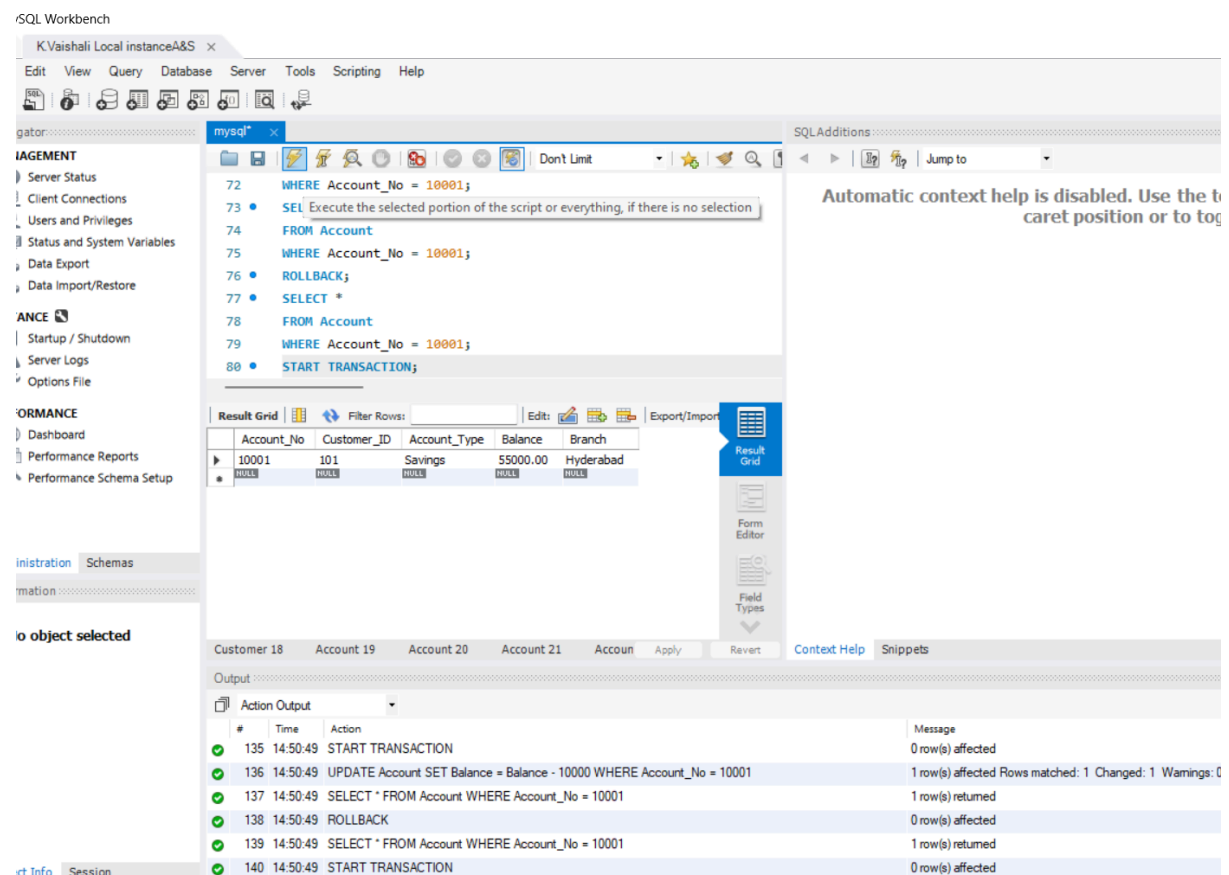
The result grid shows the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad

The output window shows the following messages:

#	Time	Action	Message
111	14:43:47	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
112	14:43:47	START TRANSACTION	0 row(s) affected
113	14:46:26	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
114	14:47:55	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
115	14:48:47	ROLLBACK	0 row(s) affected
116	14:50:00	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned

STEP-19 START TRANSACTION



The screenshot shows the MySQL Workbench interface. The query window contains the following SQL code:

```
72 WHERE Account_No = 10001;
73 SELECT *
74 FROM Account
75 WHERE Account_No = 10001;
76 ROLLBACK;
77 SELECT *
78 FROM Account
79 WHERE Account_No = 10001;
80 START TRANSACTION;
```

The result grid shows the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad

The output window shows the following messages:

#	Time	Action	Message
135	14:50:49	START TRANSACTION	0 row(s) affected
136	14:50:49	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
137	14:50:49	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
138	14:50:49	ROLLBACK	0 row(s) affected
139	14:50:49	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
140	14:50:49	START TRANSACTION	0 row(s) affected

STEP-20 FIRST UPDATE

SQL Editor:

```
75 WHERE Account_No = 10001;
76 ROLLBACK;
77 SELECT *
78 FROM Account
79 WHERE Account_No = 10001;
80 START TRANSACTION;
81 UPDATE Account
82 SET Balance = Balance + 5000
83 WHERE Account_No = 10001;
```

Result Grid:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Output:

#	Time	Action	Message
✓ 160	14:52:05	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 161	14:52:05	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
✓ 162	14:52:05	ROLLBACK	0 row(s) affected
✓ 163	14:52:05	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
✓ 164	14:52:05	START TRANSACTION	0 row(s) affected
✓ 165	14:52:05	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

STEP-21 CREATE SAVEPOINT

SQL Editor:

```
76 ROLLBACK;
77 SELECT *
78 FROM Account
79 WHERE Account_No = 10001;
80 START TRANSACTION;
81 UPDATE Account
82 SET Balance = Balance + 5000
83 WHERE Account_No = 10001;
84 SAVEPOINT Deposit1;
```

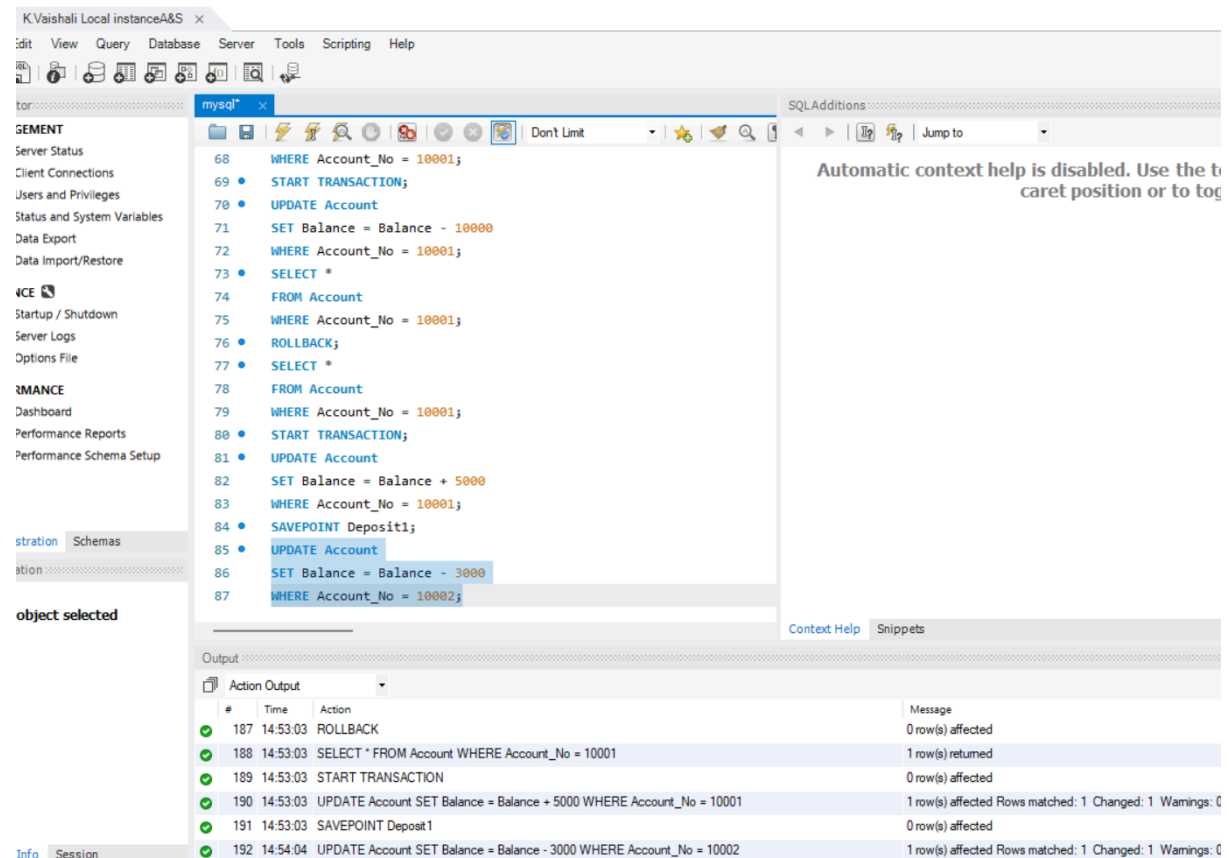
Result Grid:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Output:

#	Time	Action	Message
✓ 186	14:53:03	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
✓ 187	14:53:03	ROLLBACK	0 row(s) affected
✓ 188	14:53:03	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
✓ 189	14:53:03	START TRANSACTION	0 row(s) affected
✓ 190	14:53:03	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 191	14:53:03	SAVEPOINT Deposit1	0 row(s) affected

STEP-22 SECOND UPDATE

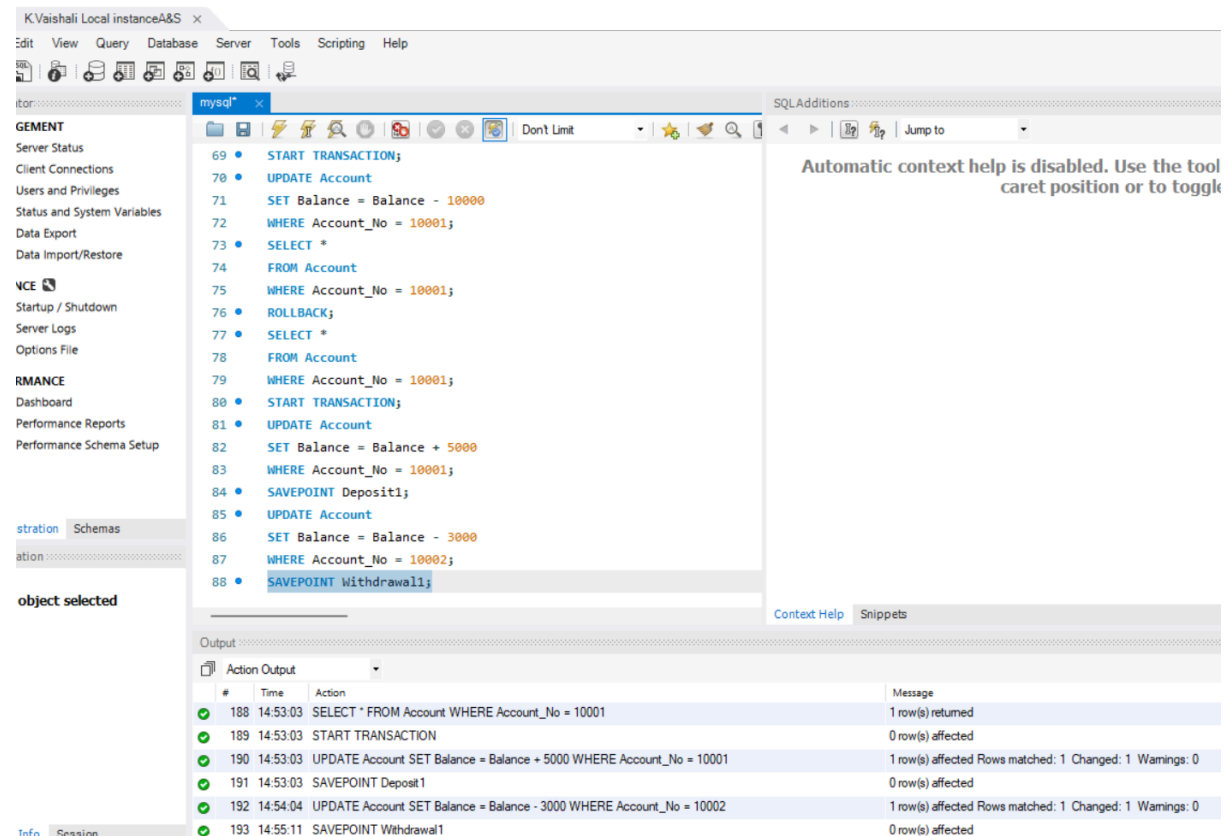


Automatic context help is disabled. Use the tool bar icon to toggle context help on or off.

```
68 WHERE Account_No = 10001;
69 START TRANSACTION;
70 UPDATE Account
71 SET Balance = Balance - 10000
72 WHERE Account_No = 10001;
73 SELECT *
74 FROM Account
75 WHERE Account_No = 10001;
76 ROLLBACK;
77 SELECT *
78 FROM Account
79 WHERE Account_No = 10001;
80 START TRANSACTION;
81 UPDATE Account
82 SET Balance = Balance + 5000
83 WHERE Account_No = 10001;
84 SAVEPOINT Deposit1;
85 UPDATE Account
86 SET Balance = Balance - 3000
87 WHERE Account_No = 10002;
```

#	Time	Action	Message
✓ 187	14:53:03	ROLLBACK	0 row(s) affected
✓ 188	14:53:03	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
✓ 189	14:53:03	START TRANSACTION	0 row(s) affected
✓ 190	14:53:03	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 191	14:53:03	SAVEPOINT Deposit1	0 row(s) affected
✓ 192	14:54:04	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

STEP-23 CREATE SECOND SAVEPOINT



Automatic context help is disabled. Use the tool bar icon to toggle context help on or off.

```
69 START TRANSACTION;
70 UPDATE Account
71 SET Balance = Balance - 10000
72 WHERE Account_No = 10001;
73 SELECT *
74 FROM Account
75 WHERE Account_No = 10001;
76 ROLLBACK;
77 SELECT *
78 FROM Account
79 WHERE Account_No = 10001;
80 START TRANSACTION;
81 UPDATE Account
82 SET Balance = Balance + 5000
83 WHERE Account_No = 10001;
84 SAVEPOINT Deposit1;
85 UPDATE Account
86 SET Balance = Balance - 3000
87 WHERE Account_No = 10002;
88 SAVEPOINT Withdrawal1;
```

#	Time	Action	Message
✓ 188	14:53:03	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
✓ 189	14:53:03	START TRANSACTION	0 row(s) affected
✓ 190	14:53:03	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 191	14:53:03	SAVEPOINT Deposit1	0 row(s) affected
✓ 192	14:54:04	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
✓ 193	14:55:11	SAVEPOINT Withdrawal1	0 row(s) affected

STEP-24 THIRD UPDATE

Automatic context help is disabled. Use the to caret position or to toggle context help.

```

72 WHERE Account_No = 10001;
73 SELECT *
74 FROM Account
75 WHERE Account_No = 10001;
76 ROLLBACK;
77 SELECT *
78 FROM Account
79 WHERE Account_No = 10001;
80 START TRANSACTION;
81 UPDATE Account
82 SET Balance = Balance + 5000
83 WHERE Account_No = 10001;
84 SAVEPOINT Deposit1;
85 UPDATE Account
86 SET Balance = Balance - 3000
87 WHERE Account_No = 10002;
88 SAVEPOINT Withdrawal1;
89 UPDATE Account
90 SET Balance = Balance + 10000
91 WHERE Account_No = 10003;

```

#	Time	Action	Message
189	14:53:03	START TRANSACTION	0 row(s) affected
190	14:53:03	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
191	14:53:03	SAVEPOINT Deposit1	0 row(s) affected
192	14:54:04	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
193	14:55:11	SAVEPOINT Withdrawal1	0 row(s) affected
194	14:56:10	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10003	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

STEP-25 ROLLBACK TO SAVEPOINT

Automatic context help is disabled. Use the to caret position or to toggle context help.

```

73 SELECT *
74 FROM Account
75 WHERE Account_No = 10001;
76 ROLLBACK;
77 SELECT *
78 FROM Account
79 WHERE Account_No = 10001;
80 START TRANSACTION;
81 UPDATE Account
82 SET Balance = Balance + 5000
83 WHERE Account_No = 10001;
84 SAVEPOINT Deposit1;
85 UPDATE Account
86 SET Balance = Balance - 3000
87 WHERE Account_No = 10002;
88 SAVEPOINT Withdrawal1;
89 UPDATE Account
90 SET Balance = Balance + 10000
91 WHERE Account_No = 10003;
92 ROLLBACK TO SAVEPOINT Withdrawal1;

```

#	Time	Action	Message
190	14:53:03	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
191	14:53:03	SAVEPOINT Deposit1	0 row(s) affected
192	14:54:04	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
193	14:55:11	SAVEPOINT Withdrawal1	0 row(s) affected
194	14:56:10	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10003	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
195	14:57:07	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected

STEP-26 COMMIT

Automatic context help is disabled. Use the tooltip caret position or to toggle

Customer 36 Account 37 Account 38 Account 39 Account Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message
221	14:58:42	SAVEPOINT Deposit1	0 row(s) affected
222	14:58:42	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
223	14:58:42	SAVEPOINT Withdrawal1	0 row(s) affected
224	14:58:42	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10003	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
225	14:58:42	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected
226	14:58:42	COMMIT	0 row(s) affected

STEP-27 CHECK ACCOUNTS

Automatic context help is disabled. Use the tooltip caret position or to toggle

Account 42 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message
222	14:58:42	UPDATE Account SET Balance = Balance - 3000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
223	14:58:42	SAVEPOINT Withdrawal1	0 row(s) affected
224	14:58:42	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10003	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
225	14:58:42	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected
226	14:58:42	COMMIT	0 row(s) affected
227	14:59:48	SELECT * FROM Account	5 row(s) returned

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	60000.00	Hyderabad
10002	102	Savings	72000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad

STEP-28 TRANSFER MONEY

K Vaishali Local instanceA&S x

Edit View Query Database Server Tools Scripting Help

mysql*

SQLAdditions

Automatic context help is disabled. Use the to caret position or to togg

```

76 • ROLLBACK;
77 • SELECT *
78 FROM Account
79 WHERE Account_No = 10001;
80 • START TRANSACTION;
81 • UPDATE Account
82 SET Balance = Balance + 5000
83 WHERE Account_No = 10001;
84 • SAVEPOINT Deposit1;
85 • UPDATE Account
86 SET Balance = Balance - 3000
87 WHERE Account_No = 10002;
88 • SAVEPOINT Withdrawal1;
89 • UPDATE Account
90 SET Balance = Balance + 10000
91 WHERE Account_No = 10003;
92 • ROLLBACK TO SAVEPOINT Withdrawal1;
93 • COMMIT;
94 • SELECT * FROM Account;
95 • START TRANSACTION;

```

Output

Action Output

#	Time	Action	Message
223	14:58:42	SAVEPOINT Withdrawal1	0 row(s) affected
224	14:58:42	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10003	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
225	14:58:42	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected
226	14:58:42	COMMIT	0 row(s) affected
227	14:59:48	SELECT * FROM Account	5 row(s) returned
228	15:01:12	START TRANSACTION	0 row(s) affected

Info Session

STEP-29 DEDUCT FROM SENDER

K Vaishali Local instanceA&S x

Edit View Query Database Server Tools Scripting Help

mysql*

SQLAdditions

Automatic context help is disabled. Use the to caret position or to togg

```

79 WHERE Account_No = 10001;
80 • START TRANSACTION;
81 • UPDATE Account
82 SET Balance = Balance + 5000
83 WHERE Account_No = 10001;
84 • SAVEPOINT Deposit1;
85 • UPDATE Account
86 SET Balance = Balance - 3000
87 WHERE Account_No = 10002;
88 • SAVEPOINT Withdrawal1;
89 • UPDATE Account
90 SET Balance = Balance + 10000
91 WHERE Account_No = 10003;
92 • ROLLBACK TO SAVEPOINT Withdrawal1;
93 • COMMIT;
94 • SELECT * FROM Account;
95 • START TRANSACTION;
96 • UPDATE Account
97 SET Balance = Balance - 10000
98 WHERE Account_No = 10001;

```

Output

Action Output

#	Time	Action	Message
224	14:58:42	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10003	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
225	14:58:42	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected
226	14:58:42	COMMIT	0 row(s) affected
227	14:59:48	SELECT * FROM Account	5 row(s) returned
228	15:01:12	START TRANSACTION	0 row(s) affected
229	15:02:38	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

Info Session

STEP-30 ADD TO RECEIVER

SQL script for Step 30:

```
82 SET Balance = Balance + 5000
83 WHERE Account_No = 10001;
84 SAVEPOINT Deposit1;
85 UPDATE Account
86 SET Balance = Balance - 3000
87 WHERE Account_No = 10002;
88 SAVEPOINT Withdrawal1;
89 UPDATE Account
90 SET Balance = Balance + 10000
91 WHERE Account_No = 10003;
92 ROLLBACK TO SAVEPOINT Withdrawal1;
93 COMMIT;
94 SELECT * FROM Account;
95 START TRANSACTION;
96 UPDATE Account
97 SET Balance = Balance - 10000
98 WHERE Account_No = 10001;
99 UPDATE Account
100 SET Balance = Balance + 10000
101 WHERE Account_No = 10002;
```

Output:

#	Time	Action	Message
225	14:58:42	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected
226	14:58:42	COMMIT	0 row(s) affected
227	14:59:48	SELECT * FROM Account	5 row(s) returned
228	15:01:12	START TRANSACTION	0 row(s) affected
229	15:02:38	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
230	15:03:57	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

STEP-31 CHECK BOTH ACCOUNTS

SQL script for Step 31:

```
96 UPDATE Account
97 SET Balance = Balance - 10000
98 WHERE Account_No = 10001;
99 UPDATE Account
100 SET Balance = Balance + 10000
101 WHERE Account_No = 10002;
102 SELECT *
103 FROM Account
104 WHERE Account_No IN (10001, 10002);
```

Result Grid:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	82000.00	Vijayawada

Output:

#	Time	Action	Message
226	14:58:42	COMMIT	0 row(s) affected
227	14:59:48	SELECT * FROM Account	5 row(s) returned
228	15:01:12	START TRANSACTION	0 row(s) affected
229	15:02:38	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
230	15:03:57	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
231	15:05:12	SELECT * FROM Account WHERE Account_No IN (10001, 10002)	2 row(s) returned

STEP-32 COMMIT TRANSFER

K.Vaishali Local instanceA&S x

Edit View Query Database Server Tools Scripting Help

mysql*

```

86 SET Balance = Balance - 3000
87 WHERE Account_No = 10002;
88 SAVEPOINT Withdrawal1;
89 UPDATE Account
90 SET Balance = Balance + 10000
91 WHERE Account_No = 10003;
92 ROLLBACK TO SAVEPOINT Withdrawal1;
93 COMMIT;
94 SELECT * FROM Account;
95 START TRANSACTION;
96 UPDATE Account
97 SET Balance = Balance - 10000
98 WHERE Account_No = 10001;
99 UPDATE Account
100 SET Balance = Balance + 10000
101 WHERE Account_No = 10002;
102 SELECT *
103 FROM Account
104 WHERE Account_No IN (10001, 10002);
105 COMMIT;

```

Automatic context help is disabled. Use the tool caret position or to toggle

Output

#	Time	Action	Message
227	14:59:48	SELECT * FROM Account	5 row(s) returned
228	15:01:12	START TRANSACTION	0 row(s) affected
229	15:02:38	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
230	15:03:57	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
231	15:05:12	SELECT * FROM Account WHERE Account_No IN (10001, 10002)	2 row(s) returned
232	15:06:10	COMMIT	0 row(s) affected

Info Session

STEP-33 FINAL BALANCES

K.Vaishali Local instanceA&S x

Edit View Query Database Server Tools Scripting Help

mysql*

```

102 SELECT *
103 FROM Account
104 WHERE Account_No IN (10001, 10002);
105 COMMIT;
106 SELECT
107     Account_No,
108     Balance
109 FROM Account
110 WHERE Account_No IN (10001, 10002);

```

Automatic context help is disabled. Use the tool caret position or to toggle

Result Grid

Account_No	Balance
10001	50000.00
10002	82000.00

Output

#	Time	Action	Message
228	15:01:12	START TRANSACTION	0 row(s) affected
229	15:02:38	UPDATE Account SET Balance = Balance - 10000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
230	15:03:57	UPDATE Account SET Balance = Balance + 10000 WHERE Account_No = 10002	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0
231	15:05:12	SELECT * FROM Account WHERE Account_No IN (10001, 10002)	2 row(s) returned
232	15:06:10	COMMIT	0 row(s) affected
233	15:07:14	SELECT Account_No, Balance FROM Account WHERE Account_No IN (10001, 10002)	2 row(s) returned

Info Session

